

Endpoint unboundedness for the Riesz-potential operator in arXiv:2001.09677

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Source Question

Hernandez–Semenov–Tradacete [1] consider, for $0 < \alpha, \lambda < 1$, a closed Ahlfors α -regular set $\Omega_\alpha \subset (0, 1)$, equipped with $\mu = \mathcal{H}^\alpha|_{\Omega_\alpha}$, and the Riesz-potential operator

$$T_\lambda f(t) = \int_0^1 |t - u|^{-\lambda} f(u) du, \quad t \in \Omega_\alpha.$$

Their Theorem 10 proves boundedness and strict singular non-compactness on the open line segment

$$\frac{1}{q} = \frac{1}{\alpha} \left(\frac{1}{p} - 1 + \lambda \right), \quad 1 - \lambda < \frac{1}{p} < \frac{1 - \lambda}{1 - \alpha}.$$

Remark 11 asks whether the endpoint operator is bounded and, if it is bounded, whether it is strictly singular.

$k_0 \in \mathbb{N}$, we have $B_k = (t_0 - \frac{1}{2k}, t_0 + \frac{1}{2k}) \subset (0, 1)$ for every $k \geq k_0$. Let

$$f_k = \frac{\chi_{B_k}}{\mu(B_k)^{\frac{1}{p}}} = k^{\frac{1}{p}} \chi_{B_k}.$$

Then, for $t \in \Omega_s$ we have

$$Tf_k(t) = k^{\frac{1}{p}} \int_{B_k} |t - u|^{-\lambda} du.$$

Let us see that the set $(T_\lambda f_k)_{k=k_0}^\infty$ is not uniformly q -integrable. Indeed, let $B'_k = B_k \cap \Omega_\alpha$. For $u \in B_k$ and $t \in B'_k$, we have $|t - u| \leq \frac{1}{k}$, hence

$$Tf_k(t) \geq k^{\lambda + \frac{1}{p} - 1}.$$

Therefore, using that $\mathcal{H}^\alpha$ is Ahlfors α -regular there exists $c_\alpha > 0$ such that

$$\|(T_\lambda f_k) \chi_{B'_k}\|_q \geq \|k^{\lambda + \frac{1}{p} - 1} \chi_{B'_k}\|_q = k^{\lambda + \frac{1}{p} - 1} \mathcal{H}^\alpha(B'_k)^{\frac{1}{q}} \geq c_\alpha > 0.$$

Since this holds for every $k \geq k_0$, it follows that $(Tf_k)_{k=k_0}^\infty$ is not uniformly q -integrable, and so $T \notin K(L_p(0, 1), L_q(\Omega_\alpha))$ as claimed (cf. [21, p. 49]). $\square$

Remark 11. The above proof leaves open the question whether the operator $T_\lambda : L_{\frac{1-s}{1-\lambda}}(0, 1) \rightarrow L_{\frac{1-\alpha}{1-\lambda}}(\Omega_\alpha)$ is actually bounded. Even in case it were, we do not know if it would be strictly singular. However, it can be deduced from Proposition

Remark 1. *The printed Remark 11 uses an undeclared symbol s in the first exponent. The endpoint of Theorem 10, and the following sentence of the same remark, identify the coherent Banach-space endpoint as*

$$p = q = \frac{1 - \alpha}{1 - \lambda}.$$

This is a Banach exponent precisely when $\alpha < \lambda$, and this is the case treated below.

Theorem

Theorem 1. *Let $0 < \alpha < \lambda < 1$, let $\Omega_\alpha \subset (0, 1)$ be closed and Ahlfors α -regular, and let $\mu = \mathcal{H}^\alpha|_{\Omega_\alpha}$. Put*

$$p = \frac{1 - \alpha}{1 - \lambda}.$$

Then T_λ is not bounded from $L_p(0, 1)$ to $L_p(\Omega_\alpha, \mu)$. Consequently, the endpoint asked about in Remark 11 is outside $L(T_\lambda)$, so the strict-singularity follow-up is vacuous.

Proof Intuition

At the endpoint, the adjoint potential generated by an Ahlfors-regular piece of Ω_α has exactly the critical weak $L^{p'}$ distribution near the fractal set. Ahlfors regularity gives two competing powers: the potential grows like $r^{\alpha-\lambda}$, while the Lebesgue measure of the r -neighborhood of the set is at least a constant times $r^{1-\alpha}$. Since

$$p' = \frac{1 - \alpha}{\lambda - \alpha},$$

these powers cancel in the strong $L^{p'}$ integral, leaving a logarithmic divergence. Thus the adjoint cannot be bounded.

Proof

Let $p' = \frac{p}{p-1}$. Since

$$p = \frac{1 - \alpha}{1 - \lambda}, \quad p - 1 = \frac{\lambda - \alpha}{1 - \lambda},$$

we have

$$p' = \frac{1 - \alpha}{\lambda - \alpha}.$$

Assume for contradiction that

$$T_\lambda : L_p(0, 1) \longrightarrow L_p(\Omega_\alpha, \mu)$$

is bounded. Then its Banach-space adjoint is bounded

$$T_\lambda^* : L_{p'}(\Omega_\alpha, \mu) \longrightarrow L_{p'}(0, 1),$$

and for $g \in L_{p'}(\Omega_\alpha, \mu)$,

$$T_\lambda^* g(x) = \int_{\Omega_\alpha} |x - t|^{-\lambda} g(t) d\mu(t) \quad \text{for a.e. } x \in (0, 1).$$

For nonnegative simple g , this formula follows by applying Fubini to truncated kernels and then using monotone convergence; boundedness would force the displayed potential to belong to $L_{p'}(0, 1)$.

Choose $t_0 \in \Omega_\alpha$ and $R > 0$ so small that $(t_0 - R, t_0 + R) \subset (0, 1)$. Set

$$E = \Omega_\alpha \cap (t_0 - R/2, t_0 + R/2), \quad E_0 = \Omega_\alpha \cap (t_0 - R/4, t_0 + R/4), \quad g = \mathbf{1}_E.$$

Then $g \in L_{p'}(\Omega_\alpha, \mu)$. We will show that

$$P(x) := T_\lambda^* g(x) = \int_E |x - t|^{-\lambda} d\mu(t)$$

does not belong to $L_{p'}(0, 1)$.

Let $0 < r < R/16$, and consider the r -neighborhood

$$N_r = \{x \in (0, 1) : \text{dist}(x, E_0) < r\}.$$

If $x \in N_r$, choose $y \in E_0$ with $|x - y| < r$. For every $t \in \Omega_\alpha \cap (y - r, y + r)$ we have $|x - t| < 2r$. Since $\Omega_\alpha \cap (y - r, y + r) \subset E$, Ω_α is Ahlfors α -regular, and $r < R/16$,

$$\mu(\Omega_\alpha \cap (y - r, y + r)) \geq cr^\alpha,$$

where $c > 0$ depends only on the regularity constants. Hence

$$P(x) \geq (2r)^{-\lambda} \mu(\Omega_\alpha \cap (y - r, y + r)) \geq c_1 r^{\alpha - \lambda} \quad (x \in N_r).$$

We also need a lower bound for the Lebesgue measure of N_r . Take a maximal $2r$ -separated subset $\{y_1, \dots, y_m\}$ of E_0 . The intervals $(y_i - r, y_i + r)$ are disjoint and contained in N_r , while maximality and the upper Ahlfors estimate give

$$\mu(E_0) \leq \sum_{i=1}^m \mu(\Omega_\alpha \cap (y_i - 2r, y_i + 2r)) \leq Cmr^\alpha.$$

The left-hand side is a positive constant depending on R . Therefore $m \geq c_R r^{-\alpha}$, and

$$|N_r| \geq \sum_{i=1}^m |(y_i - r, y_i + r)| \geq c_2 r^{1 - \alpha}.$$

Write $a = \lambda - \alpha > 0$. The previous two estimates imply the distribution lower bound

$$|\{x \in (0, 1) : P(x) \geq c_1 r^{-a}\}| \geq c_2 r^{1 - \alpha} \quad (0 < r < R/16).$$

Since $p' = (1 - \alpha)/a$, this is exactly the critical weak $L^{p'}$ decay. To see that the strong $L^{p'}$ norm diverges, take $r_n = 2^{-n} r_0$, with $r_0 < R/16$, and $\tau_n = c_1 r_n^{-a}$. By the layer-cake formula,

$$\int_0^1 P(x)^{p'} dx = p' \int_0^\infty \tau^{p'-1} |\{x : P(x) > \tau\}| d\tau.$$

For $\tau \in [\tau_n, \tau_{n+1}]$,

$$|\{P > \tau\}| \geq |\{P > \tau_{n+1}\}| \geq c r_{n+1}^{1 - \alpha} = c' \tau_{n+1}^{-p'}.$$

Thus each interval contributes at least

$$c' p' \tau_{n+1}^{-p'} \int_{\tau_n}^{\tau_{n+1}} \tau^{p'-1} d\tau = c'' \left(1 - \left(\frac{\tau_n}{\tau_{n+1}}\right)^{p'}\right),$$

which is bounded below by a positive constant independent of n . Summing over n gives

$$\int_0^1 P(x)^{p'} dx = \infty.$$

Therefore $T_\lambda^* g \notin L_{p'}(0, 1)$, contradicting boundedness of the adjoint. Hence T_λ is not bounded at the endpoint.

Verification Notes

The proof uses only the two-sided Ahlfors regularity of Ω_α and elementary duality. It does not rely on any endpoint trace theorem. The condition $\alpha < \lambda$ is exactly what makes the endpoint exponent $(1 - \alpha)/(1 - \lambda)$ larger than 1; if $\lambda = \alpha$ the endpoint is $p = 1$, and if $\lambda < \alpha$ it is below the Banach range.

The source already proves noncompactness for the open segment. The present argument shows that the missing endpoint is not merely noncompact; it is not bounded at all.

Novelty Check

Local index searches for 2001.09677, **strictly singular non compact, Riesz potential, endpoint**, and related L_p -characteristic terms found no prior packet for this source question. Web searches for exact endpoint phrases, the source title together with **T_lambda**, and Adams/Riesz trace-inequality keywords found broad trace-inequality literature, including classical endpoint caveats for $p = q$, but no direct answer to Remark 11 of Hernandez–Semenov–Tradacete.

Human Review Recommendation

Recommended for review as a likely valid full answer to the endpoint boundedness question in Remark 11, interpreting the remark’s undeclared s as the Ahlfors exponent α dictated by Theorem 10.

References

- [1] F. L. Hernandez, E. M. Semenov, and P. Tradacete, *Strictly singular non-compact operators between L_p spaces*, arXiv:2001.09677.